3) (10, 10)

from A → 1 pile
from B → 1 pile
from A & B → 1, 1 pile

you win if you make both pile after every move Opponent moves

eventually

Opponent (1, 0)

(0, 0) → You

$\boxed{\text{Second player wins}}$

10 –
9 –
8 –
7 –
6 –
5 –
4 –
3 –
2 –
1

A → Opponent
B → You

(10, 10) → (9, 10) A'
(9, 9) → (8, 9)
B'
(8, 8) → (7, 8)
B' :
:
(1, 1) → (0, 1)
(0, 0) → A' → You win

(b) payoff → getting some portion of stag > hare > Nothing

part 1. $\geq m$ hunters catching stag
Sab stag choose karein
↳ Equilibrium → koi strategy nahi badlega as $\boxed{U(\frac{1}{n}\text{stag}) > U(\text{hare})}$

2. Sab hare choose karein
ek bhi stag ke piche pada to fail hojayega
hence wo strategy nahi badlega
↳ equilibrium

$\boxed{\text{equilibrium = All stag and all Hare}}$

part-2 → $(m \leq k \leq n)$ $\boxed{\frac{1}{k}\text{stag} > \text{hare}}$

$\boxed{\frac{1}{n}\text{stag} < \text{hare}}$

Hunter stag tab choose karega
when stag hunter kam ho
exactly k hunters

$\frac{1}{k}$ mil ra

If one changes strategy then $\frac{1}{k+1}$ milra worst then hare hence noone will change strategy

↳ Nash equilibrium

More than k hunter
share chota sab hare pasand karege
So equilibrium nahi

All hare → Nash equilibrium

Game theory

Q2(a) good is provided if k people contribute to it.

priority → ① good bana paise nhi diye.
② public good provided and gave money
③ public good didn't provided & didn't gave money
④ public good not provided & contributed

① >k contributed for public good

(n−k) → nahi karenge contribute phir bhi aajayega hence he will not give and not contribute hence No equillibrium exists.

② =k contributed for public good

k needed

k contributed → $\boxed{\text{Exactly k contributer = Nash eq}}$ hence

↳ sab chahenge good aajaye to withdraw nhi karega koi

& jo nhi kr ra wo karega to loss hai use hence he will also not change his strategy.

③ <k

→ koi contribute nahi karsha

koi strategy change nahi karega couz of loss hence it is also NE.

→ kuch kar rahe but good nhi aarha hence strategy changes and No NE here.